APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Acids, Bases & Natural Indicators

SUBJECT: General Science (SCI101)

CLASS & SECTION: Class 7 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 91 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Test turmeric, red cabbage, and litmus paper reactions with lemon juice, vinegar, soap, and lime water.

1. Executive Overview & Problem Context

Chemical substances display distinct acidic or alkaline characteristics based on their hydrogen ion (H z) or hydroxide ion (OH {) dissociation in aqueous solutions. Acid-base indicators are weak organic chromophores that undergo conformational color shifts at specific pH thresholds.

2. Core Analysis, Solutions & Systematic Working

- **Chemical Properties of Acids & Bases:**

  Acids possess a sour taste, pH < 7, release H z(aq) ions, and turn blue litmus red (e.g. Citric acid in lemon, Acetic acid in vinegar, Hydrochloric acid in gastric juice). Bases feel slippery/soapy, pH > 7, release OH {(aq) ions, and turn red litmus blue (e.g. Sodium hydroxide in soap, Calcium hydroxide in lime water).

- **Natural Chromophore: Turmeric Extract (Curcumin):**

  Curcumin remains vibrant yellow in neutral and acidic solutions (lemon juice, vinegar, tap water). In basic solutions (baking soda slurry, liquid soap, detergent water), curcumin undergoes structural deprotonation above pH 8.5, shifting to a deep reddish-brown tone.

- **Anthocyanin Extraction: Red Cabbage Indicator:**

  Red cabbage contains water-soluble anthocyanin pigments serving as a universal natural pH indicator: turns bright red in strong acids (pH 1-2), pink in weak acids (pH 3-5), purple in neutral solutions (pH 7), emerald green in weak bases (pH 8-10), and yellow in strong bases (pH > 11).

- **Neutralization Reactions & Everyday Applications:**

  Acid + Base -> Salt + Water + Heat. HCl + NaOH -> NaCl + H ,O. Practical applications: Magnesium hydroxide (Milk of Magnesia) neutralizing excess gastric acidity; Calamine lotion (Zinc carbonate) soothing formic acid injected by ant stings; Slaked lime treating acidic agricultural soils.

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3. Detailed Data Matrix & Systematic Findings

Tested Household Sample	Expected Chemical	Turmeric Color	Litmus Result	pH Type
Fresh Lemon Juice	Citric Acid (C †H ˆO ‡)	Stays Yellow	Blue Litmus -> Red	Acidic (pH ~2.2)
White Kitchen Vinegar	Dilute Acetic Acid	Stays Yellow	Blue Litmus -> Red	Acidic (pH ~2.8)
Laundry Detergent Slurry	Sodium Alkyl Sulfate	Deep Red-Brown	Red Litmus -> Blue	Basic (pH ~10.5)
Baking Soda in Water	Sodium Bicarbonate	Reddish Orange	Red Litmus -> Blue	Mild Base (pH ~8.4)
Pure Distilled Water	H ,O (Neutral)	Stays Yellow	No Color Change	Neutral (pH 7.0)

4. Student Conclusions & Reflective Synthesis

Natural pigments offer eco-friendly, non-toxic diagnostic indicators for classifying aqueous solutions, illustrating fundamental chemical equilibrium and neutralization principles.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Analytical Depth & Research Methodology: 28 / 30

Problem Solving & Structured Working: 18 / 20

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Very good effort. Clear and organized chemical logs, accurate pH color scale recordings, and balanced neutralization equations."

APEX EVALUATION DIVISION

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